The Laurent-tail kill at $n = 7$: seven layer-0 survivors and how they die

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(notes towards locality at $n \geq 7$)

Abstract

At n external legs, the layer-0 ("Step-1") hidden-zero kill mechanism eliminates the coefficient a_M of every chord multiset M that, at some zone $Z_{r,r+2}$, omits the eliminated vertex $r+1$ and avoids the special chord $X_{r,r+2}$. Through $n = 6$ this is enough: the mechanism kills every non-triangulation. At $n = 7$, however, exactly seven non-triangulation multisets escape – a cyclic family of "fish." We show that all seven die at the next Laurent layer. For each fish M there is a zone Z at which the order X_S^{-3} Laurent expansion of $1/\prod X_c$ produces a free-variable fingerprint shared by exactly six ansatz monomials, and all five cousins of M are killed by layer 0 at a single neighbouring zone. The fingerprint equation collapses to $a_M = 0$. Verification is symbolic, performed by the companion script `cascade_kill_n7.py` at [1], with full output in `results_cascade_n7.txt`.

1 Why the fish exist: layer-0 misses seven heptagon multisets

We work with a heptagon ($n = 7$) with vertices $1, 2, \ldots, 7$ in cyclic order. Chords are pairs (i, j) with $1 \leq i < j \leq 7$ that are not polygon edges. The general planar rational ansatz of mass dimension $-2(n-3) = -8$ is

$$B \;=\; \sum_M \frac{a_M}{\prod_{c \in M} X_c},$$

where M ranges over multisets of $N := n-3 = 4$ chords. A multiset is a *triangulation* iff its four chords are pairwise non-crossing; the $\mathrm{Tr}(\phi^3)$ tree amplitude is a sum over the $C_5 = 42$ heptagon triangulations.

Zones and layer-0 kill. For each $r \in \{1, \ldots, n\}$, the zone $Z_{r,r+2}$ is the codimension-2 hidden-zero locus on which the chords incident to vertex $r+1$ become linear combinations of chords incident to r and the special chord $S = X_{r,r+2}$. Explicitly,

$$X_{r+1,k} \;=\; X_{r,k} - S \qquad (k \neq r, r+1, r+2),$$

and the bare relation $X_{r+1,r-1} = -S$. At zone $Z_{r,r+2}$, every chord is exactly one of: special (S), bare ($X_{r+1,r-1}$), non-bare substituted ($X_{r+1,k}$, with companion $X_{r,k}$ free), or plain free (every chord avoiding both r and $r+1$). Sending $S \to \infty$ and reading the leading Laurent order, linear independence of the free-variable monomials forces $a_M = 0$ whenever

(K1) M contains neither S nor the bare;

(K2) for every (companion, substitute) pair, M does not contain both members.

We write $M \in \mathcal{K}_{r,r+2}$ when (K1)+(K2) hold.

The seven layer-0 survivors at $n = 7$. A complete enumeration[1] shows that exactly seven non-triangulation multisets satisfy $M \notin \mathcal{K}_{r,r+2}$ for *every* r:

$$\begin{aligned} M_1 &= \{(1,3),(1,6),(2,4),(4,6)\}, & M_2 &= \{(1,3),(1,6),(3,5),(4,6)\}, \\ M_3 &= \{(1,3),(1,6),(3,5),(5,7)\}, & M_4 &= \{(1,3),(2,7),(3,5),(5,7)\}, \\ M_5 &= \{(1,6),(2,4),(2,7),(4,6)\}, & M_6 &= \{(2,4),(2,7),(3,5),(5,7)\}, \\ M_7 &= \{(2,4),(2,7),(4,6),(5,7)\}. \end{aligned} \tag{1}$$

They are the seven cyclic rotations of M_1 under $v \mapsto v+\sigma \bmod 7$ for $\sigma = 0, 2, 4, 6, 5, 1, 3$ respectively. Visually each is a "fish" of two crossing chords plus a containing pair of chords (cf. the survivor diagram in [1]).

Proposition 1 (Layer-0 fails on every fish)**.** *For every $i \in \{1, \ldots, 7\}$ and every $r \in \{1, \ldots, 7\}$, $M_i \notin \mathcal{K}_{r,r+2}$.*

Proof. By cyclic symmetry it suffices to do $i = 1$. The vertices used by M_1 are $\{1, 2, 3, 4, 6\}$; vertices 5 and 7 are unused. For $M_1 \in \mathcal{K}_{r,r+2}$, condition (K1) requires the special $X_{r,r+2} \notin M_1$ and the bare $X_{r+1,r-1} \notin M_1$. Test each r:

- $r = 1$: special $X_{1,3} \in M_1$; (K1) fails.
- $r = 2$: special $X_{2,4} \in M_1$; (K1) fails.
- $r = 3$: special $X_{3,5} \notin M_1$, bare $X_{4,2} = (2,4) \in M_1$; (K1) fails.
- $r = 4$: special $X_{4,6} = (4,6) \in M_1$; (K1) fails.
- $r = 5$: special $X_{5,7} \notin M_1$, bare $X_{6,4} = (4,6) \in M_1$; (K1) fails.
- $r = 6$: special $X_{6,1} = (1,6) \in M_1$; (K1) fails.
- $r = 7$: special $X_{7,2} = (2,7) \notin M_1$, bare $X_{1,6} = (1,6) \in M_1$; (K1) fails.

So (K1) fails at every zone, and $M_1 \notin \mathcal{K}_{r,r+2}$ for all r. □ □

The proof shows the structural reason: each fish either uses the special or contains the bare at every cyclic zone. This is what makes the fish "unbreakable" by layer-0 – they are precisely the multisets whose chords cyclically saturate the special/bare slots.

2 The Laurent-tail kill

The mechanism. At zone $Z_{r,r+2}$ with $S = X_{r,r+2}$, the geometric-series identity

$$\frac{1}{X_{r,k} - S} = -\frac{1}{S} - \frac{X_{r,k}}{S^2} - \frac{X_{r,k}^2}{S^3} - \cdots \tag{2}$$

expands every non-bare substitute $1/X_{r+1,k}$ as a Laurent series in $1/S$. The bare $1/X_{r+1,r-1} = -1/S$ has no tail. Multiplying N such factors gives, for each ansatz monomial, a Laurent series in $1/S$ whose coefficients are rational *free-variable* monomials – monomials in the companions $\{X_{r,k}\}$ and the born-free chords (those avoiding both r and $r+1$). Algebraic independence of the free variables implies that every free-variable monomial at every order in S has total coefficient zero on $B|_{Z_{r,r+2}} = 0$.

[1] Performed by `step1_uncaught.py` in the companion repository [1]; the output is `nonlocal_n7_step1_survivors.pdf`.

Notation. A *fingerprint* is a specific rational free-variable monomial $\mathcal{F}$. The *fingerprint equation* of $(\mathcal{F}, k, r)$ is the equation that the order-$S^{-(N+k)}$ coefficient of $\mathcal{F}$ in $B|_{Z_{r,r+2}}$ equals zero. Layer k refers to subleading order k beyond layer 0.

The kill of M_1. Take $M_1 = \{(1,3),(1,6),(2,4),(4,6)\}$ at zone $Z_{5,7}$, where $S = X_{5,7}$, eliminated vertex 6, and the substitutions are

$$X_{1,6} = X_{1,5} - S, \quad X_{2,6} = X_{2,5} - S, \quad X_{3,6} = X_{3,5} - S, \quad X_{4,6} = -S. \tag{3}$$

The chords $(1,3)$ and $(2,4)$ avoid both 5 and 6, hence are free. The Laurent expansion of the fish monomial on $Z_{5,7}$ is

$$\frac{1}{X_{1,3}\,X_{2,4}\,X_{1,6}\,X_{4,6}} = \frac{1}{X_{1,3}X_{2,4}} \cdot \frac{1}{X_{1,5} - S} \cdot \frac{-1}{S}.$$

Using (2), the order S^{-3} piece is $\dfrac{X_{1,5}}{X_{1,3}X_{2,4}} \cdot \dfrac{1}{S^3}$. We isolate the fingerprint

$$\mathcal{F}_1 := \frac{X_{1,5}}{X_{1,3}X_{2,4}}.$$

Lemma 2 (Six-monomial fingerprint equation for M_1)**.** *The fingerprint $\mathcal{F}_1$ at order S^{-3} on $Z_{5,7}$ receives contributions from exactly six ansatz monomials. The fingerprint equation is*

$$\begin{aligned} 0 \;=\; &-\, a_{\{(1,3),(1,5),(1,6),(2,4)\}} \;+\; 2\, a_{\{(1,3),(1,6),(1,6),(2,4)\}} \;+\; a_{\{(1,3),(1,6),(2,4),(2,6)\}} \\ &+\, a_{\{(1,3),(1,6),(2,4),(3,6)\}} \;+\; a_{\{(1,3),(1,6),(2,4),(4,6)\}} \;-\; a_{\{(1,3),(1,6),(2,4),(5,7)\}}. \end{aligned} \tag{4}$$

Proof. A monomial of B contributes a power of $X_{1,5}$ in the numerator at order S^{-3} iff one of its chord factors is incident to vertex 6 on the substituted side – specifically iff it contains $X_{1,6}$, because $X_{1,5}$ enters only through the Laurent tail (3) of $1/X_{1,6}$. (The companions of $X_{2,6}$ and $X_{3,6}$ supply $X_{2,5}$ and $X_{3,5}$ respectively, not $X_{1,5}$.) Likewise the denominator of $\mathcal{F}_1$ requires the monomial to contain $X_{1,3}$ and $X_{2,4}$ each with multiplicity 1. A size-4 multiset thus has chord factors $\{X_{1,3}, X_{2,4}, X_{1,6}, X_a\}$ for some fourth chord X_a. Direct expansion using (2) gives each of the six contributing $X_a \in \{X_{1,5}, X_{1,6}, X_{2,6}, X_{3,6}, X_{4,6}, X_{5,7}\}$ the integer scalar shown. The full computation is performed by `fingerprint_equation` in `cascade_kill_n7.py`. □ □

Lemma 3 (All five cousins die by layer-0 at $Z_{4,6}$)**.** *Each of the five non-fish multisets in* (4) *satisfies $M \in \mathcal{K}_{4,6}$, hence its coefficient is killed by layer-0 at zone $Z_{4,6}$.*

Proof. At $Z_{4,6}$: special $X_{4,6} = (4,6)$, bare $X_{5,3} = (3,5)$, companion-substitute pairs $(X_{4,7}, X_{5,7})$, $(X_{4,1}, X_{5,1}) = (X_{1,4}, X_{1,5})$, $(X_{4,2}, X_{5,2}) = (X_{2,4}, X_{2,5})$. We check (K1)+(K2) for each cousin:

- $\{(1,3),(1,5),(1,6),(2,4)\}$: special $\notin$, bare $\notin$, pair $(X_{1,4}, X_{1,5})$ has only $X_{1,5}$, others have at most one. ✓
- $\{(1,3),(1,6),(1,6),(2,4)\}$: special $\notin$, bare $\notin$, no pair has both members. ✓
- $\{(1,3),(1,6),(2,4),(2,6)\}$: special $\notin$, bare $\notin$, no pair has both ($X_{2,4} \in$, $X_{2,5} \notin$). ✓
- $\{(1,3),(1,6),(2,4),(3,6)\}$: special $\notin$, bare $\notin$, no pair has both. ✓
- $\{(1,3),(1,6),(2,4),(5,7)\}$: special $\notin$, bare $\notin$, pair $(X_{4,7}, X_{5,7})$ has only $X_{5,7}$, pair $(X_{2,4}, X_{2,5})$ has only $X_{2,4}$. ✓

All five satisfy (K1)+(K2) at $Z_{4,6}$. □ □

Theorem 4 (Cascade kill of all seven $n = 7$ fish)**.** *For each* $i \in \{1, \ldots, 7\}$, $a_{M_i} = 0$.

Proof. For $i = 1$, substitute Lemma 3 into (4): all five cousin coefficients are zero, so the equation collapses to $+a_{M_1} = 0$. For $i = 2, \ldots, 7$, apply the cyclic shift $v \mapsto v+\sigma_i \bmod 7$ to the entire $i = 1$ argument: the recipe $(Z_{5,7}, \mathcal{F}_1)$ becomes $(Z_{5+\sigma_i,7+\sigma_i}, \mathrm{rot}_{\sigma_i}(\mathcal{F}_1))$, the six-monomial equation becomes the cyclic rotate of (4), and the cousins' $Z_{4,6}$-layer-0 kill becomes $Z_{4+\sigma_i,6+\sigma_i}$-layer-0 kill. The companion script `cascade_kill_n7.py` recomputes the rotated equation directly (rather than rotating the symbolic answer), confirming the recipe and verifying every cousin kill at the rotated zone; see `results_cascade_n7.txt`. □ □

3 Companion code and reproducibility

The script `cascade_kill_n7.py` reproduces every step of Theorem 4 symbolically. The main routines:

- `laurent_coefficient(M, r, n, k)`: Laurent expansion of $1/\prod_{c\in M} X_c$ on $Z_{r,r+2}$ to order S^{-k}, via `sympy.series` after the substitution $S \mapsto 1/u$.
- `coefficient_of_monomial(expr, target, free_vars)`: extract the scalar coefficient of a target free-variable monomial from a sum of rational monomials.
- `candidate_multisets_for_fingerprint(fp, n, N)`: fast pre-filter that enumerates only multisets whose chord factors are compatible with producing `fp` (so the search runs in seconds rather than hours).
- `fingerprint_equation(M, r, n, k, fp, pool)`: returns $[(M', \text{scalar})]$ for every multiset that contributes the fingerprint at order S^{-k} on $Z_{r,r+2}$.
- `verify_each_fish_from_scratch`: for each fish, computes its kill recipe by cyclic rotation of the M_1 recipe and *verifies it directly* by recomputing the fingerprint equation and checking that every cousin is layer-0 killable.
- `verify_all_seven_fish`: top-level driver, writes the human-readable cascade trace to `results_cascade_n7.txt`.

The output file `results_cascade_n7.txt` contains, for each fish, the kill zone and order, the fingerprint, the six-term equation, and the layer-0 kill zone of every cousin. All seven kills are confirmed.

4 Outlook

The cascade has a strikingly simple shape: *one Laurent step deeper at one zone, with prerequisites all of layer-0 type at one neighbouring zone.* The cyclic structure of (1) reflects this – each fish is the rotated image of M_1, and the kill recipe rotates with it. This raises the natural conjecture that every layer-0 survivor at $n \geq 7$ dies via a depth-1 cascade of the same shape: identify the per-fish recipe combinatorially, then verify the fingerprint equation's cousins are all layer-0 killable. The next experiment is to check this at $n = 8, 9$ using the same script with the obvious adaptations. If the depth-1 phenomenon persists, the path to a clean all-n locality proof is to reduce the all-layer fingerprint problem to a single-step combinatorial induction.

References

[1] GitHub repository *locality-proof*: https://github.com/Jony-GydeRnE/locality-proof. Files: `cascade_kill_n7.py` (verifier), `results_cascade_n7.txt` (cascade trace), `step1_uncaught.py` (layer-0 enumeration), `nonlocal_n7_step1_survivors.pdf` (the seven fish drawn).

[2] A. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory*, arXiv:2406.04234.

[3] N. Arkani-Hamed, Q. Cao, J. Dong, C. Figueiredo, S. He, *Hidden zeros for particle/string amplitudes and the unity of colored scalars, pions, and gluons*, arXiv:2312.16282.